Adaptive Iterative Learning Control-Based Rotor Position Harmonic Error Suppression Method for Sensorless PMSM Drives

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***Abstract*—Accurate rotor position estimation is important to ensure the high-performance operation of position sensorless permanent magnet synchronous motor (PMSM) drives. To suppress the estimated rotor position harmonic error (ERPHE) caused by inverter nonlinearities, flux spatial harmonics, and the current measurement error in high-frequency signal injection-based schemes, an adaptive iterative learning control-based online suppression method is proposed in this article. This method constructs a P-type iterative learning rotor position observer with the forgetting factor to obtain the tracking error and generate the compensation value in real time. The stability, the convergence, and the parameter sensitivity of this observer are analyzed in detail. To improve the suppression effect of the ERPHE for different loads and speeds, an iterative learning gain self-tuning strategy is investigated. The proposed method does not require the offline detection and the discrimination of harmonic frequencies, which has strong suppression ability for different harmonic error components. The effectiveness is verified by experiments on a 2.2-kW interior PMSM drive platform.**

***Index Terms*—Adaptive iterative learning control, high frequency signal injection (HFSI), online suppression, permanent magnet synchronous motor (PMSM), rotor position harmonic error.**

I. Introduction

PERMANENT magnet synchronous motors (PMSMs) have been widely applied in the industrial manufacturing due to the advantages of high efficiency and high power density [1]–[4]. In order to reduce costs and improve the system robustness, position sensorless control strategies for PMSMs have been investigated [5]–[8]. High-frequency signal injection (HFSI)-based methods can be used in the zero- and low-speed range for sensorless PMSM drives [9]–[12]. However, affected by inverter nonlinearities, flux spatial harmonics, the current measurement error, and other factors, there is a harmonic error with complex harmonic components in the estimated rotor position when HFSI-based methods are adopted [13]–[15]. The harmonic error makes the field-oriented control decoupling inaccurate, and causes the speed and torque to fluctuate, which even causes the sensorless system to operate abnormally.

In order to further improve the performance of sensorless operation, the estimated rotor position harmonic error (ERPHE) needs to be suppressed. Considering the causes of the ERPHE, it can be diminished by optimizing the motor structure and the control algorithm. It is an effective solution to compensate the nonideal saliency characteristics by improving the structure design [16], [17]. An HFSI-based position estimation scheme was incorporated directly into the finite element design process to improve the sensorless control performance [16]. In [17], the influence of rotor geometry on induced high-frequency (HF) currents was analyzed, and the interior PMSM (IPMSM) rotor design criteria were determined to improve the estimation accuracy. Although the optimization of motor structure can suppress the ERPHE, the comprehensive performance indicators need to be considered. Alternatively, the control algorithm optimization-based ERPHE suppression methods are flexible and practical, which can be classified into two categories: the compensation-based methods [18]–[24] and the digital filtering-based methods [15], [25]–[29].

Compensation-based methods can improve the position estimation accuracy by compensating inverter nonlinearities, the current measurement error, and the resulting ERPHE [18]–[24]. To weaken the negative influence of inverter nonlinearities, the voltage error signal was modeled and compensated based on the current polarity [19]. In [20], an HF equivalent resistance model was proposed to analyze the influence of inverter nonlinearities on HFSI. As a result, the position estimation accuracy

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was improved effectively. In [21], the influence of zero-current clamping effects on HFSI was systematically described and diminished. In [22], a voltage feedforward compensation method based on the iterative learning control algorithm was proposed to suppress the harmonics of position estimation error. This method is independent of the current polarity and the inverter model. On the other hand, the positive and negative sequence components of induced HF currents were used for correcting the error of current measurement gain [23]. In [24], the integrator output of the *d*-axis current regulator was utilized to offset the current measurement error. Consequently, these methods can effectively compensate the inherent nonideal characteristics of the motor drive system combining the causes of the ERPHE, thereby improving the position estimation accuracy. However, they are limited in the comprehensive suppression ability for complex harmonic error caused by multiple factors in HFSI-based methods.

Digital filtering-based methods can effectively suppress the ERPHE by directly eliminating harmonic error components during the rotor position estimation [15], [25]–[29]. The adaptive filtering strategies have been proven to be an effective harmonics detection and suppression method [25]–[29]. In [27], an adaptive linear neural-network-based filter was proposed to eliminate the high-order harmonic components of the estimated back electromotive force, which can diminish the position estimation harmonic ripples. Similarly, two synchronous frequency-extract filters were applied to decrease the estimated back electromotive force harmonic error adaptively [28]. In [15], the second order generalized integrator and the notch filters were used to diminish the sixth and the second harmonic components of the ERPHE, respectively, in the HFSI-based method. Research shows that digital filtering-based methods can effectively suppress the harmonic components without considering the causes of the ERPHE. However, these methods increase the algorithm complexity and the harmonic frequency information should be known in advance. Additionally, the harmonic frequencies of the ERPHE are close to the fundamental frequency at low-speed operation, which affects the filtering performance.

This article proposes an adaptive iterative learning control (AILC)-based online ERPHE suppression method for sensorless PMSM drives, contributing to improving the mitigation ability for complex harmonic error components when using HFSI-based methods. By analyzing the comprehensive causes of the ERPHE due to nonideal factors, a P-type iterative learning rotor position observer (PILO) with the forgetting factor is proposed to diminish the ERPHE in real time. Based on the analysis results of the stability, the convergence, and the parameter sensitivity of the PILO in the discrete time domain, an iterative learning gain self-tuning strategy is investigated. In this strategy, the iterative learning gains are adaptively adjusted by comparing the constructed mean error evaluation function with the expected error, which improves the universality and the adaptability for different loads and speeds. This method does not need to distinguish harmonic error frequencies and can avoid complex adaptive filtering algorithms. The effectiveness of the proposed suppression method is verified by experiments on a 2.2-kW IPMSM platform.

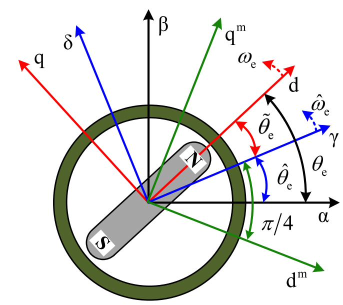

Fig. 1. Reference frames of PMSM.

II. Analysis of Position Estimation Error for HFSI-Based PMSM Drives

A. Rotor Position Estimation in HFSI-Based Methods

The reference frames used for analyzing the ERPHE in HFSI-based sensorless PMSM drives are shown in Fig. 1. The α–β, the d–q, the γ–δ, and the d^{m}–q^{m} frames denote the stationary, the rotary, the estimated, and the measured reference frames, respectively. The measured frame lags behind the estimated frame by 45°. θ_{e}, $\hat{\theta}_{\mathrm{e}}$, and $\tilde{\theta}_{\mathrm{e}}$ denote the actual rotor position, the estimated rotor position, and the rotor position estimation error, respectively. ω_{e} and $\hat{\omega}_{\mathrm{e}}$ are the actual and the estimated motor electrical angular speeds, respectively. In the following, the HF square-wave voltage injection-based sensorless PMSM drive is adopted to analyze the ERPHE. In order to amplify the coefficient of the position error information and improve the performance of error signal equivalence and position estimation, HF signals are processed in the d^{m}–q^{m} frame.

In HFSI-based methods, since the back electromotive force is relatively low when the motor operates at low speed and the resistance is much smaller than the d–q axis inductive reactance, the effect of the variation of resistance and flux is negligible. The HF mathematical model of PMSM in the γ–δ frame can be expressed as [11]

$$\begin{bmatrix} u_{\gamma\mathrm{h}} \\ u_{\delta\mathrm{h}} \end{bmatrix} = \begin{bmatrix} U_{\mathrm{i}}\xi_{\mathrm{sq}}^{T_{\mathrm{i}}}(t) \\ 0 \end{bmatrix} = \boldsymbol{T}(\tilde{\theta}_{\mathrm{e}}) \begin{bmatrix} L_{\mathrm{dh}} & 0 \\ 0 & L_{\mathrm{qh}} \end{bmatrix} \boldsymbol{T}^{-1}(\tilde{\theta}_{\mathrm{e}}) \cdot p \begin{bmatrix} i_{\gamma\mathrm{h}} \\ i_{\delta\mathrm{h}} \end{bmatrix} \tag{1}$$

where $u_{\gamma\mathrm{h}}$, $u_{\delta\mathrm{h}}$, $i_{\gamma\mathrm{h}}$, and $i_{\delta\mathrm{h}}$ are the γ–δ axis HF voltages and currents, respectively, L_{dh} and L_{qh} are the d–q axis incremental self-inductances, respectively, p is the differential operator, U_{i} and T_{i} are the amplitude and the period of the injected HF voltage, respectively, $\xi_{\mathrm{sq}}^{T_{\mathrm{i}}}(t)$ is a unit-amplitude square-wave signal with a period of T_{i}, and $\boldsymbol{T}(\cdot)$ is the coordinate transformation matrix, which can be expressed as [30]

$$\xi_{\mathrm{sq}}^{T_{\mathrm{i}}}(t) = \begin{cases} 1 & t_{\mathrm{rem}} \in (0, T_{\mathrm{i}}/2] \\ -1 & t_{\mathrm{rem}} \in (T_{\mathrm{i}}/2, T_{\mathrm{i}}] \end{cases} \tag{2}$$

$$\boldsymbol{T}(\cdot) = \begin{bmatrix} \cos(\cdot) & -\sin(\cdot) \\ \sin(\cdot) & \cos(\cdot) \end{bmatrix} \tag{3}$$

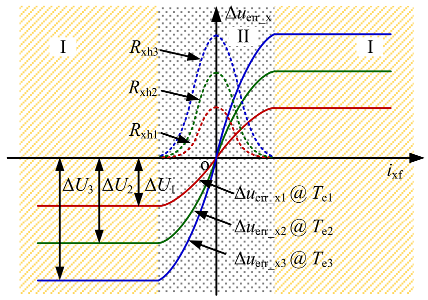

Fig. 2. Relationship of the voltage error with the current and the load.

where t_{rem} denotes the remainder of t divided by T_{i}.

The induced HF currents in the d^{m}–q^{m} frame are obtained as

$$\begin{aligned}\begin{bmatrix} i_{\mathrm{d^m h}} \\ i_{\mathrm{q^m h}} \end{bmatrix} &= \frac{U_{\mathrm{i}} T_{\mathrm{i}} \xi_{\mathrm{tri}}^{T_{\mathrm{i}}}(t)}{4\sqrt{2}(\Sigma L_{\mathrm{h}}^2 - \Delta L_{\mathrm{h}}^2)} \begin{bmatrix} \Sigma L_{\mathrm{h}} - \Delta L_{\mathrm{h}}(\cos 2\tilde{\theta}_{\mathrm{e}} - \sin 2\tilde{\theta}_{\mathrm{e}}) \\ \Sigma L_{\mathrm{h}} - \Delta L_{\mathrm{h}}(\cos 2\tilde{\theta}_{\mathrm{e}} + \sin 2\tilde{\theta}_{\mathrm{e}}) \end{bmatrix} \\ &= \xi_{\mathrm{tri}}^{T_{\mathrm{i}}}(t) \begin{bmatrix} I_{\mathrm{d^m h}} \\ I_{\mathrm{q^m h}} \end{bmatrix} \end{aligned} \tag{4}$$

where $i_{\mathrm{d^m h}}$, $i_{\mathrm{q^m h}}$, $I_{\mathrm{d^m h}}$, and $I_{\mathrm{q^m h}}$ are the d^{m}–q^{m} axis HF currents and their amplitudes, respectively, ΣL_{h} and ΔL_{h} are the average and the difference inductances, respectively, which are defined as $\Sigma L_{\mathrm{h}} = (L_{\mathrm{dh}} + L_{\mathrm{qh}})/2$ and $\Delta L_{\mathrm{h}} = (L_{\mathrm{dh}} - L_{\mathrm{qh}})/2$, and $\xi_{\mathrm{tri}}^{T_{\mathrm{i}}}(t)$ is a unit-amplitude triangular-wave signal with a period of T_{i}, which can be expressed as

$$\xi_{\mathrm{tri}}^{T_{\mathrm{i}}}(t) = \begin{cases} -1 + 4t_{\mathrm{rem}}/T_{\mathrm{i}} & t_{\mathrm{rem}} \in (0, T_{\mathrm{i}}/2] \\ 3 - 4t_{\mathrm{rem}}/T_{\mathrm{i}} & t_{\mathrm{rem}} \in (T_{\mathrm{i}}/2, T_{\mathrm{i}}] \end{cases}. \tag{5}$$

According to (4), the amplitude difference of the d^{m}–q^{m} axis HF currents contains the rotor position error information as

$$\tilde{\theta}_{\mathrm{e}} \approx \frac{\sqrt{2}\left(\Sigma L_{\mathrm{h}}^2 - \Delta L_{\mathrm{h}}^2\right)}{U_{\mathrm{i}} T_{\mathrm{i}} \Delta L_{\mathrm{h}}} \left(I_{\mathrm{d^m h}} - I_{\mathrm{q^m h}}\right) = \lambda \left(I_{\mathrm{d^m h}} - I_{\mathrm{q^m h}}\right). \tag{6}$$

Therefore, the rotor position can be estimated according to (6) in combination with a phase-locked loop position observer.

B. Analysis of Estimated Rotor Position Harmonic Error

Affected by inverter nonlinearities, there is a voltage error between the actual and the theoretical injected HF voltages, as shown in Fig. 2. $\Delta u_{\mathrm{err\_x}}$, i_{xf}, and R_{xh} denote the x-phase voltage error, fundamental current, and HF equivalent resistance. ΔU is the saturation value. T_{e} is the electromagnetic torque. The relationship between $\Delta u_{\mathrm{err\_x}}$ and i_{xf} can be divided into the saturation region (region I) and the nonlinear region (region II) [20]. In region I, the influence of the constant voltage error on the HF currents extraction can be ignored. In region II, affected by R_{xh}, when three-phase HF currents alternate rapidly, the alternating voltage error causes the $(1\pm6k)$th harmonic ripples in the three-phase and the α–β axis HF current amplitudes (or envelopes) [27], which can be expressed as

$$\begin{cases} i_{\alpha\mathrm{h}} = \xi_{\mathrm{tri}}^{T_{\mathrm{i}}}(t) \Big\{ I_{\alpha\mathrm{h1}} \cos\left(\hat{\omega}_{\mathrm{e}} t + \varphi_{\alpha\mathrm{h1}}\right) \\ \qquad + \sum\limits_{k=1} I_{\alpha\mathrm{h1}\pm6\mathrm{k}} \cos\left(\left(1 \pm 6k\right)\hat{\omega}_{\mathrm{e}} t + \varphi_{\alpha\mathrm{h1}\pm6\mathrm{k}}\right) \Big\} \\ i_{\beta\mathrm{h}} = \xi_{\mathrm{tri}}^{T_{\mathrm{i}}}(t) \Big\{ I_{\beta\mathrm{h1}} \sin\left(\hat{\omega}_{\mathrm{e}} t + \varphi_{\beta\mathrm{h1}}\right) \\ \qquad + \sum\limits_{k=1} I_{\beta\mathrm{h1}\pm6\mathrm{k}} \sin\left(\left(1 \pm 6k\right)\hat{\omega}_{\mathrm{e}} t + \varphi_{\beta\mathrm{h1}\pm6\mathrm{k}}\right) \Big\} \end{cases} \tag{7}$$

where $i_{\alpha\mathrm{h}}$ and $i_{\beta\mathrm{h}}$ are the α–β axis HF currents, $I_{\alpha\mathrm{h1}}$, $I_{\beta\mathrm{h1}}$, $\varphi_{\alpha\mathrm{h1}}$, $\varphi_{\beta\mathrm{h1}}$, $I_{\alpha\mathrm{h1}\pm6\mathrm{k}}$, $I_{\beta\mathrm{h1}\pm6\mathrm{k}}$, $\varphi_{\alpha\mathrm{h1}\pm6\mathrm{k}}$, and $\varphi_{\beta\mathrm{h1}\pm6\mathrm{k}}$ are the amplitude and the phase of the fundamental component and the $(1\pm6k)$th harmonics of the α–β axis HF current envelopes, respectively.

According to (7), there are the $(6k)$th harmonic ripples in the d^{m}–q^{m} axis HF current envelopes after the coordinate transformation [27], which can be expressed as

$$\begin{aligned}&\begin{bmatrix} i_{\mathrm{d^m h}} \\ i_{\mathrm{q^m h}} \end{bmatrix} \\ &= \xi_{\mathrm{tri}}^{T_{\mathrm{i}}}(t) \begin{bmatrix} I_{\mathrm{d^m h1}} + \sum\limits_{k=1} I_{\mathrm{d^m h6k}} \cos\left(6k\hat{\omega}_{\mathrm{e}} t + \varphi_{\mathrm{d^m h6k}}\right) \\ I_{\mathrm{q^m h1}} + \sum\limits_{k=1} I_{\mathrm{q^m h6k}} \sin\left(6k\hat{\omega}_{\mathrm{e}} t + \varphi_{\mathrm{q^m h6k}}\right) \end{bmatrix}\end{aligned} \tag{8}$$

where $I_{\mathrm{d^m h1}}$, $I_{\mathrm{q^m h1}}$, $I_{\mathrm{d^m h6k}}$, $I_{\mathrm{q^m h6k}}$, $\varphi_{\mathrm{d^m h6k}}$, and $\varphi_{\mathrm{q^m h6k}}$ are the dc component, the amplitude and the phase of the $(6k)$th harmonics of the d^{m}–q^{m} axis HF current envelopes, respectively.

In practical situations, there are inevitable offset and scaling error during the current measurement, which can also cause harmonic ripples in the HF current amplitudes. It is assumed that the a- and b-phase measured currents contain the offset and the scaling error, which can be expressed as

$$\begin{cases} i'_{\mathrm{a}} = k_{\mathrm{a}} i_{\mathrm{a}} + \sigma_{\mathrm{a}} = k_{\mathrm{a}}(i_{\mathrm{af}} + i_{\mathrm{ah}}) + \sigma_{\mathrm{a}} = k_{\mathrm{a}} i_{\mathrm{af}} + \sigma_{\mathrm{a}} + k_{\mathrm{a}} i_{\mathrm{ah}} \\ i'_{\mathrm{b}} = k_{\mathrm{b}} i_{\mathrm{b}} + \sigma_{\mathrm{b}} = k_{\mathrm{b}}(i_{\mathrm{bf}} + i_{\mathrm{bh}}) + \sigma_{\mathrm{b}} = k_{\mathrm{b}} i_{\mathrm{bf}} + \sigma_{\mathrm{b}} + k_{\mathrm{b}} i_{\mathrm{bh}} \end{cases} \tag{9}$$

where k_{a}, k_{b}, σ_{a}, and σ_{b} are the a- and b-phase scale factors and offset values, respectively, the superscript represents the measured value.

Then, the measured values of the d^{m}–q^{m} axis HF currents can be obtained as

$$\begin{aligned}\begin{bmatrix} i'_{\mathrm{d^m h}} \\ i'_{\mathrm{q^m h}} \end{bmatrix} &= \boldsymbol{T}^{-1}\left(\hat{\theta}_{\mathrm{e}} - \frac{\pi}{4}\right) \begin{bmatrix} k_{\mathrm{a}} i_{\mathrm{ah}} \\ \frac{\sqrt{3}}{3}\left(k_{\mathrm{a}} i_{\mathrm{ah}} + 2k_{\mathrm{b}} i_{\mathrm{bh}}\right) \end{bmatrix} \\ &= k_{\mathrm{a}} \begin{bmatrix} i_{\mathrm{d^m h}} \\ i_{\mathrm{q^m h}} \end{bmatrix} + \frac{\sqrt{6}k_{\mathrm{a}}}{3} \begin{bmatrix} i_{\mathrm{bh}}\left(\frac{k_{\mathrm{b}}}{k_{\mathrm{a}}} - 1\right)\left(\sin\hat{\theta}_{\mathrm{e}} - \cos\hat{\theta}_{\mathrm{e}}\right) \\ i_{\mathrm{bh}}\left(\frac{k_{\mathrm{b}}}{k_{\mathrm{a}}} - 1\right)\left(\sin\hat{\theta}_{\mathrm{e}} + \cos\hat{\theta}_{\mathrm{e}}\right) \end{bmatrix}. \end{aligned} \tag{10}$$

According to (10), the measured d^{m}–q^{m} axis HF currents are related to the scale factors. If $k_{\mathrm{a}} \neq k_{\mathrm{b}}$, additional harmonic ripples exist in the HF current amplitudes. Considering inverter nonlinearities and the current measurement error, substituting

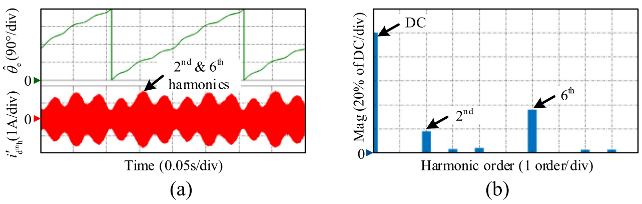

Fig. 3. Simulation results of the d^{m}–q^{m} axis HF current amplitudes distortion. (a) d^{m}-axis HF current and the estimated rotor position. (b) FFT analysis of the d^{m}-axis HF current amplitude.

(8) and i_{bh} into (10), (10) can be rewritten as

$$i_{\mathrm{bh}} = \xi_{\mathrm{tri}}^{T_{\mathrm{i}}}(t)\left\{ I_{\mathrm{bh1}}\cos\left(\hat{\omega}_{\mathrm{e}}t + \varphi_{\mathrm{bh1}}\right) + \sum_{k=1} I_{\mathrm{bh1}\pm 6\mathrm{k}}\cos\left((1\pm 6k)\,\hat{\omega}_{\mathrm{e}}t + \varphi_{\mathrm{bh1}\pm 6\mathrm{k}}\right)\right\} \tag{11}$$

$$\begin{bmatrix} i'_{\mathrm{d^m h}} \\ i'_{\mathrm{q^m h}} \end{bmatrix} = k_{\mathrm{a}}\xi_{\mathrm{tri}}^{T_{\mathrm{i}}}(t) \times \left\{ \begin{bmatrix} I_{\mathrm{d^m h1}} + \sum\limits_{k=1} I_{\mathrm{d^m h6k}}\cos(6k\hat{\omega}_{\mathrm{e}}t + \varphi_{\mathrm{d^m h6k}}) \\ I_{\mathrm{q^m h1}} + \sum\limits_{k=1} I_{\mathrm{q^m h6k}}\sin(6k\hat{\omega}_{\mathrm{e}}t + \varphi_{\mathrm{q^m h6k}}) \end{bmatrix} + \begin{bmatrix} I_{\mathrm{d^m h2}}\cos\left(2\hat{\omega}_{\mathrm{e}}t + \varphi_{\mathrm{d^m h2}}\right) \\ I_{\mathrm{q^m h2}}\sin\left(2\hat{\omega}_{\mathrm{e}}t + \varphi_{\mathrm{q^m h2}}\right) \end{bmatrix} + \begin{bmatrix} \sum\limits_{k=1} I_{\mathrm{d^m h2\pm 6k}}\cos\left((2\pm 6k)\,\hat{\omega}_{\mathrm{e}}t + \varphi_{\mathrm{d^m h2\pm 6k}}\right) \\ \sum\limits_{k=1} I_{\mathrm{q^m h2\pm 6k}}\sin\left((2\pm 6k)\,\hat{\omega}_{\mathrm{e}}t + \varphi_{\mathrm{q^m h2\pm 6k}}\right) \end{bmatrix} \right\} \tag{12}$$

where I_{bh1}, φ_{bh1}, $I_{\mathrm{bh1}\pm 6\mathrm{k}}$, and $\varphi_{\mathrm{bh1}\pm 6\mathrm{k}}$ are the amplitude and the phase of the fundamental component and the ($1\pm 6k$)th harmonics of the b-phase HF current envelope, respectively, $I_{\mathrm{d^m h2}}$, $I_{\mathrm{q^m h2}}$, $\varphi_{\mathrm{d^m h2}}$, $\varphi_{\mathrm{q^m h2}}$, $I_{\mathrm{d^m h2\pm 6k}}$, $I_{\mathrm{q^m h2\pm 6k}}$, $\varphi_{\mathrm{d^m h2\pm 6k}}$, and $\varphi_{\mathrm{q^m h2\pm 6k}}$ are the amplitude and the phase of the second harmonics and the ($2\pm 6k$)th harmonics of the measured d^{m}–q^{m} axis HF current envelopes, respectively.

According to (12), the measured d^{m}–q^{m} axis HF current amplitudes contain complex harmonic components. The second and the ($6k$)th harmonics are the main components. However, the ($2\pm 6k$)th harmonic ripples are caused by the current measurement scaling error acting on the ($1\pm 6k$)th harmonics, which have smaller amplitudes and can be ignored. Fig. 3 shows the simulation waveforms of the d^{m}–q^{m} axis HF current amplitudes distortion and the fast Fourier transform (FFT), verifying the theoretical analysis. Only considering the first two terms in (12) and substituting them into (6), the rotor position error can be obtained as

$$\tilde{\theta}_{\mathrm{e}} \approx \lambda k_{\mathrm{a}}\left\{ (I_{\mathrm{d^m h1}} - I_{\mathrm{q^m h1}}) + \left(\sum_{k=1} I_{\mathrm{d^m h6k}}\cos\left(6k\hat{\omega}_{\mathrm{e}}t + \varphi_{\mathrm{d^m h6k}}\right) - \sum_{k=1} I_{\mathrm{q^m h6k}}\sin\left(6k\hat{\omega}_{\mathrm{e}}t + \varphi_{\mathrm{q^m h6k}}\right) + \sum_{k=1} I_{\mathrm{d^m h2}}\cos\left(2\hat{\omega}_{\mathrm{e}}t + \varphi_{\mathrm{d^m h2}}\right) - \sum_{k=1} I_{\mathrm{q^m h2}}\sin\left(2\hat{\omega}_{\mathrm{e}}t + \varphi_{\mathrm{q^m h2}}\right)\right)\right\} = \lambda k_{\mathrm{a}}\cdot\varepsilon \tag{13}$$

where ε is the equivalent position error signal input into the observer, which contains the second and the ($6k$)th harmonic ripples, causing the ERPHE.

III. Proposed AILC-Based Rotor Position Harmonic Error Suppression Method

A. Proposed P-Type Iterative Learning Rotor Position Observer With Forgetting Factor

Fig. 4 shows the block diagram of sensorless controlled PMSM drive with the proposed AILC-based online ERPHE suppression method. As an interesting control strategy, iterative learning control has a satisfactory suppression effect on the periodic disturbance [31]. As shown in Fig. 4, a PILO with the forgetting factor is proposed to suppress the ERPHE. In the PILO, Λ is the forgetting factor. $\Gamma(t)$ and $\Phi(t)$ are time-varying iterative learning gains, which can be adaptively adjusted according to the load and the speed variation. E is the expected error value. Combined with the motor drive system, the iterative learning law can be illustrated as

$$\begin{cases} \Delta I_{\mathrm{c,k+1}}(t) = (1-\Lambda)\,\Delta I_{\mathrm{c,k}}(t) + \Phi(t)\,\tilde{\omega}_{\mathrm{e,k}}(t) \\ \qquad\qquad + \Gamma(t)\,\tilde{\omega}_{\mathrm{e,k+1}}(t) \\ \tilde{\omega}_{\mathrm{e,k+1}}(t) = \Delta\omega_{\mathrm{e}}{}^{*}(t) - \Delta\hat{\omega}_{\mathrm{e,k+1}}(t) \\ \Delta\hat{\omega}_{\mathrm{e,k+1}}(t) = \mathrm{M}(t)\,\Delta I_{\mathrm{c,k+1}}(t) \end{cases} \tag{14}$$

where $\Delta I_{\mathrm{c,k+1}}(t)$, $\Delta\hat{\omega}_{\mathrm{e,k+1}}(t)$, and $\tilde{\omega}_{\mathrm{e,k+1}}(t)$ are the (k+1)th control signal (compensation value of HF current amplitudes difference), output signal (estimated speed ripple), and error signal (estimated speed harmonic error) of the iterative learning system, respectively. $k = 0, 1, 2\ldots$ is the iteration number. $\Delta\omega_{\mathrm{e}}{}^{*}(t)$ is the reference speed ripple, which can be defined as $\Delta\omega_{\mathrm{e}}{}^{*}(t) = \omega_{\mathrm{e}}{}^{*}(t) - \hat{\omega}_{\mathrm{e}}(t)$ in the sensorless drive system. $\mathrm{M}(t)$ is the transfer function between the output and the control signals of the iterative learning system. Since the control signal is a fluctuating ac signal, and the integration time constant and k_{i} are relatively small, the effect of the integral operation can be approximately neutralized at steady state. Then, $\mathrm{M}(t)$ can be defined as k_{p}. In the following, based on Fig. 4 and (14), the AILC-based ERPHE suppression process is introduced.

a) Set the initial values of Λ, $\Gamma(t)$, $\Phi(t)$, $\omega_{\mathrm{e}}{}^{*}(t)$, k, $\Delta I_{\mathrm{c}}(t)$, and E according to the control system requirements.
b) Extract the equivalent position error signal ε by processing the d^{m}–q^{m} axis HF currents.
c) Compensate the control signal $\Delta I_{\mathrm{c,k}}(t)$ to ε and generate $\hat{\omega}_{\mathrm{e,k}}(t)$ through the PILO. Compare $\omega_{\mathrm{e}}{}^{*}(t)$ with $\hat{\omega}_{\mathrm{e,k}}(t)$ to obtain and memorize the tracking error $\tilde{\omega}_{\mathrm{e,k}}(t)$.
d) Self-tune the iterative learning gains $\Gamma(t)$ and $\Phi(t)$ based on the tracking error $\tilde{\omega}_{\mathrm{e}}(t)$ and the expected error E.
e) Generate and memorize the control signal $\Delta I_{\mathrm{c,k+1}}(t)$ by using $\Delta I_{\mathrm{c,k}}(t)$, $\tilde{\omega}_{\mathrm{e,k}}(t)$, $\Gamma(t)$, and $\Phi(t)$ based on (14).

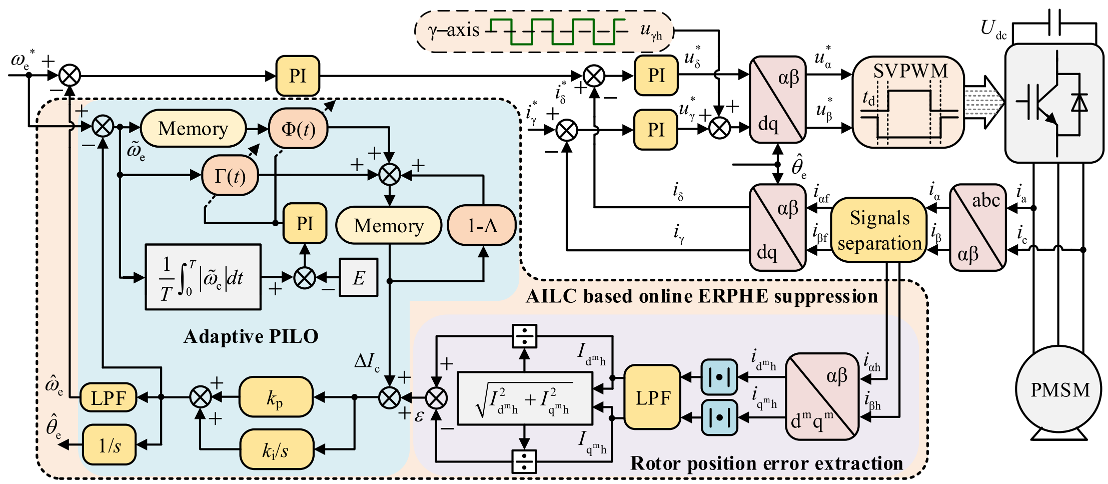

Fig. 4. Block diagram of sensorless controlled PMSM drive with the proposed AILC-based online ERPHE suppression method.

Repeat steps b) - f) to keep the tracking error at the expected value.

The initial values of $\Gamma(t)$, $\Phi(t)$, k, and $\Delta I_c(t)$ can be set to 0, $\omega_e{}^*(t)$ is the reference speed of system operation. The selection rule of Λ is relatively broad since the adaptive PILO system can adjust $\Gamma(t)$ and $\Phi(t)$ based on different Λ to ensure that the ERPHE suppression effect can reach the expected standard. The range of Λ is usually 0∼1 and it is set to 0.5 in this article. E serves as a standard to adjust the iterative learning gain by comparing with the mean error evaluation function and it can be set according to the rated speed, such as 1% rated speed in this article. Λ and E can be slightly adjusted according to the control system requirements.

B. Analysis of Convergence

The convergence of the PILO is discussed as follows. Based on (14), the (k+1)th tracking error of the iterative learning system can be expressed as

$$\begin{aligned}\tilde{\omega}_{\mathrm{e,k+1}}(t) &= \Delta\omega_e{}^*(t) - \Delta\hat{\omega}_{\mathrm{e,k+1}}(t) = \Delta\omega_e{}^*(t) - \mathrm{M}(t)\cdot\\ &\quad ((1-\Lambda)\Delta I_{\mathrm{c,k}}(t) + \Phi(t)\tilde{\omega}_{\mathrm{e,k}}(t)\\ &\quad +\Gamma(t)\tilde{\omega}_{\mathrm{e,k+1}}(t))\\ &= (1-\Lambda-\mathrm{M}(t)\Phi(t))\tilde{\omega}_{\mathrm{e,k}}(t) + \Lambda\Delta\omega_e{}^*(t)\\ &\quad -\mathrm{M}(t)\Gamma(t)\tilde{\omega}_{\mathrm{e,k+1}}(t). \end{aligned} \tag{15}$$

Then, the relationship between the (k+1)th and the kth tracking error can be obtained as

$$\begin{aligned}\tilde{\omega}_{\mathrm{e,k+1}}(t) &= \frac{(1-\Lambda-\mathrm{M}(t)\Phi(t))\tilde{\omega}_{\mathrm{e,k}}(t)}{(1+\mathrm{M}(t)\Gamma(t))} + \frac{\Lambda\Delta\omega_e{}^*(t)}{(1+\mathrm{M}(t)\Gamma(t))}\\ &= \Psi(t)\tilde{\omega}_{\mathrm{e,k}}(t) + \mathrm{H}(t). \end{aligned} \tag{16}$$

According to (16), the condition for the PILO system convergence can be illustrated as

$$|\Psi(t)| = \left|\frac{(1-\Lambda-\mathrm{M}(t)\Phi(t))}{(1+\mathrm{M}(t)\Gamma(t))}\right| < 1. \tag{17}$$

Taking the infinite norm on both sides of (16) and combining the norm inequality, the following can be obtained as:

$$\begin{aligned}\|\tilde{\omega}_{\mathrm{e,k+1}}(t)\|_\infty &= \|\Psi(t)\tilde{\omega}_{\mathrm{e,k}}(t) + \mathrm{H}(t)\|_\infty\\ &\le \|\Psi(t)\tilde{\omega}_{\mathrm{e,k}}(t)\|_\infty + \|\mathrm{H}(t)\|_\infty\\ &= |\Psi(t)|\,\|\tilde{\omega}_{\mathrm{e,k}}(t)\|_\infty + |\mathrm{H}(t)|_{\max}\\ &\le |\Psi(t)|^2\|\tilde{\omega}_{\mathrm{e,k-1}}(t)\|_\infty + (1+|\Psi(t)|)\,|\mathrm{H}(t)|_{\max} \le \ldots\ldots\\ &\le |\Psi(t)|^{k+1}\|\tilde{\omega}_{\mathrm{e,0}}(t)\|_\infty + \left(1+\ldots\ldots+|\Psi(t)|^{k}\right)|\mathrm{H}(t)|_{\max}\\ &= |\Psi(t)|^{k+1}\|\tilde{\omega}_{\mathrm{e,0}}(t)\|_\infty + \frac{1-|\Psi(t)|^{k+1}}{1-|\Psi(t)|}|\mathrm{H}(t)|_{\max}. \end{aligned} \tag{18}$$

According to (18), when the iteration number k approaches infinity, the following can be obtained as:

$$\begin{aligned}\lim_{k\to\infty}\|\tilde{\omega}_{\mathrm{e,k+1}}(t)\|_\infty &= \lim_{k\to\infty}\Bigg(|\Psi(t)|^{k+1}\|\tilde{\omega}_{\mathrm{e,0}}(t)\|_\infty\\ &\qquad + \frac{1-|\Psi(t)|^{k+1}}{1-|\Psi(t)|}|\mathrm{H}(t)|_{\max}\Bigg)\\ &= \frac{|\mathrm{H}(t)|_{\max}}{1-|\Psi(t)|}. \end{aligned} \tag{19}$$

According to (19), the tracking error has a limit value, which satisfies the convergence condition. The limit value depends on the values of Λ, Γ, and Φ. Fig. 5 shows the relationship of the tracking error limit value versus different Λ, Γ, and Φ. As can be seen from Fig. 5, when Λ is constant, the error limit value gradually decreases as Γ and Φ increase. However, there is a convergent minimum. When the error limit value reaches the convergent minimum, it would not be affected by Γ and Φ. Actually, the convergent minimum depends on the value of Λ. When Γ and Φ are constant, the smaller Λ is, the lower both the tracking error limit value and the convergent minimum are. Specially, when Λ is zero, $|\mathrm{H}(t)|$ is zero and the tracking error

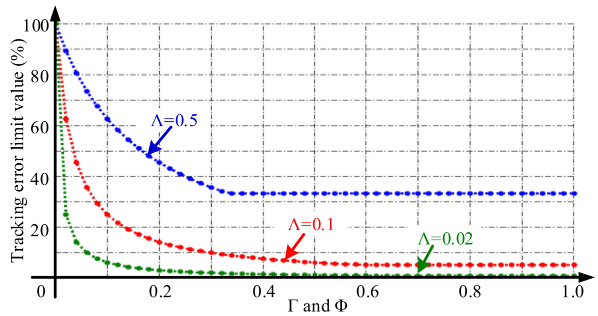

Fig. 5. Relationship of tracking error limit value versus different Λ, Γ, and Φ.

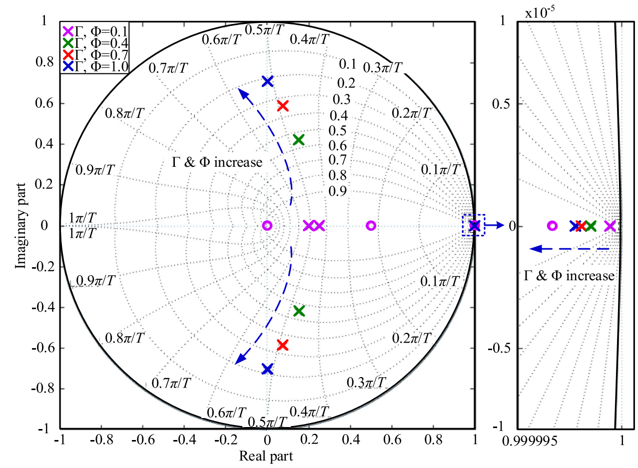

Fig. 6. Pole-zero diagram of the PILO.

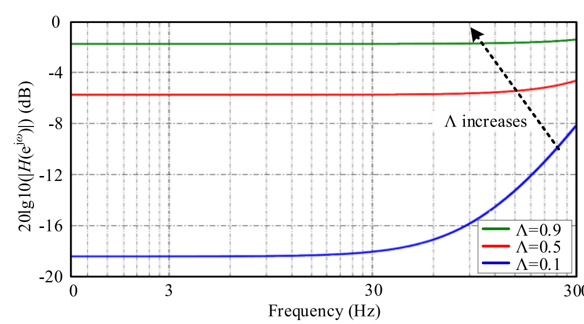

Fig. 7. Magnitude-frequency characteristic of the PILO versus different Λ ($\Gamma = \Phi = 0.4$).

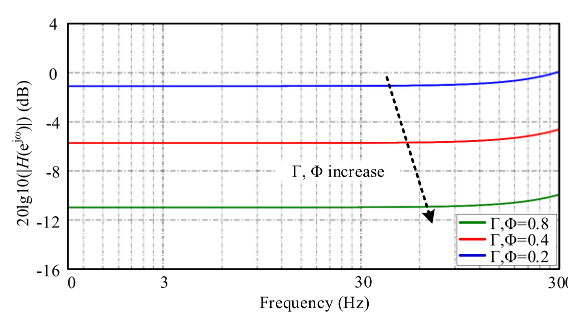

Fig. 8. Magnitude-frequency characteristic of the PILO versus different Γ and Φ ($\Lambda = 0.5$).

can also converge to zero according to (19). Correspondingly, Fig. 5 can also obtain the same analysis result.

Nevertheless, the introduction of the forgetting factor can weaken the superposition of aperiodic disturbance signals in the system. Considering the estimated speed and position harmonic error only needs to meet the acceptable range, hence, the value of Λ should not be too small.

C. Analysis of Stability and Parameter Sensitivity

In the PILO shown in Fig. 4, the memory part is equivalent to the one-beat pulsewidth modulation (PWM) period delay part, which can be expressed as z^{-1}. Then, the system closed-loop transfer function in the discrete time domain can be expressed as (20), where $G_{\mathrm{PI}}(z)$ and $G_{\mathrm{ILC}}(z)$ are the transfer function of the PI term and the ILC term, respectively, T_{s} is the PWM period, k_{p} and k_{i} are the proportional and the integral gains of the PILO, respectively.

In order to analyze the stability of the system, the pole-zero diagram of the PILO can be obtained according to (20), as shown in Fig. 6. Based on the experimental motor drive system, when T_{s}, Λ, k_{p}, and k_{i} are set to 1/6000 s, 0.5, 3, and 0.01, respectively, as can be seen from Fig. 6, the poles of the closed-loop transfer function are within the unit circle when Γ and Φ change, verifying the stability. As Γ and Φ increase, the poles are close to the unit circle. Moreover, due to the existence of aperiodic disturbance signals, the selection of Γ and Φ should not be too large.

In order to further analyze the influence of parameters on the ERPHE suppression effect, according to (20), the frequency response characteristics of the PILO system should be analyzed. Fig. 7 shows the magnitude-frequency characteristic of the PILO versus different Λ. When Λ is selected to be a larger value, such as 0.9, the attenuation effect of the system is reduced, which weakens the ERPHE suppression performance. As Λ decreases, although the attenuation effect becomes satisfactory, the suppression ability for aperiodic disturbance signals becomes weak. The analysis results are consistent with those obtained in Fig. 5.

Fig. 8 shows magnitude-frequency characteristic of the PILO versus different Γ and Φ. It can be seen that when Γ and Φ are selected to be larger values, such as 0.8, the ERPHE suppression effect is satisfactory. Larger Γ and Φ may amplify the effect of aperiodic disturbance signals, thereby reducing the stability. As

$$G_{\mathrm{c}}(z) = \frac{G_{\mathrm{PI}}(z)}{1+G_{\mathrm{PI}}(z)\cdot G_{\mathrm{ILC}}(z)} = \frac{k_{\mathrm{p}}z^3 + (k_{\mathrm{p}}\Lambda + k_{\mathrm{i}}T_{\mathrm{s}} - 2k_{\mathrm{p}})\,z^2 + (k_{\mathrm{i}}\Lambda T_{\mathrm{s}} - k_{\mathrm{p}}\Lambda - k_{\mathrm{i}}T_{\mathrm{s}} + k_{\mathrm{p}})\,z}{z^3 + (k_{\mathrm{p}}\Gamma + \Lambda - 2)\,z^2 + (k_{\mathrm{p}}\Phi - k_{\mathrm{p}}\Gamma + k_{\mathrm{i}}\Gamma T_{\mathrm{s}} + 1 - \Lambda)\,z + k_{\mathrm{i}}\Phi T_{\mathrm{s}} - k_{\mathrm{p}}\Phi} \tag{20}$$

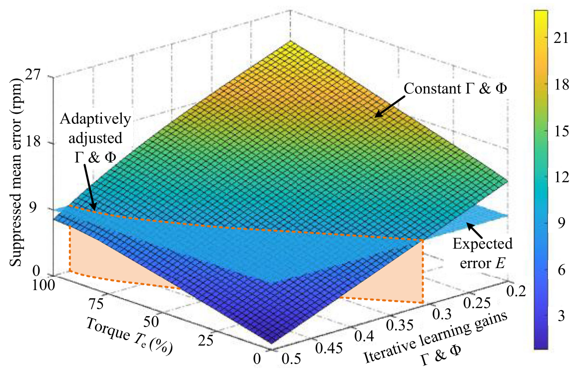

Fig. 9. Suppressed mean error versus different Γ, Φ, and loads.

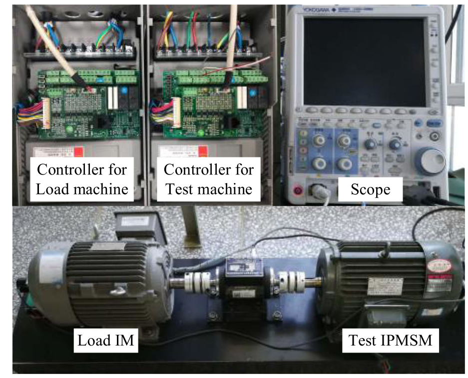

Fig. 10. Experimental platform with a 2.2-kW IPMSM.

Γ and Φ decrease, the ERPHE suppression ability becomes weak and it is difficult to achieve the expected effect.

As for the transient position error, when Γ and Φ are constant, the larger k_p and k_i are, the larger the transient position error is. Similarly, when k_p and k_i are constant, the larger Γ and Φ are, the larger the transient position error is. Hence, in order to improve the suppression effect of the ERPHE at the transient state, the selection of k_p and k_i should not be too large. The proposed adaptive PILO does not need to predetermine Γ and Φ, which effectively solves the issue of difficult Γ and Φ selection for different loads and speeds. By comparing the constructed mean error evaluation function with the expected error, Γ and Φ can be tuned adaptively using a PI regulator in the same way simultaneously, as shown in the following, improving the mitigation ability of the ERPHE:

$$\Gamma(t) = \Phi(t) = k_{pa}\left(\frac{1}{T}\int_0^T |\tilde{\omega}_e(t)|dt - E\right) + k_{ia}\int_0^T \left(\frac{1}{T}\int_0^T |\tilde{\omega}_e(t)|dt - E\right)dt \tag{21}$$

where k_{pa} and k_{ia} are the proportional and the integral gains of the self-tuning PI regulator, respectively. k_{pa} and k_{ia} can be designed as $k_{pa} = \omega_a^* \cdot T_s$ and $k_{ia} = \omega_a^* \cdot \Lambda$ according to the self-tuning loop parameters, where ω_a^* is the expected bandwidth of the self-tuning loop and it can be selected based on the PWM carrier frequency. In this article, ω_a^*, T_s, and Λ are 100 Hz, 1/6000 s, and 0.5, respectively. Therefore, k_{pa} and k_{ia} are designed as 0.017 and 50, respectively. Smaller k_{pa} and k_{ia} may affect the self-tuning speed of the iterative learning gains and the dynamic adjustment performance. When larger values are selected, the relatively severe disturbance signals may deteriorate the system stability although the self-tuning speed is improved. Hence, the parameters of the self-tuning PI regulator can be slightly adjusted according to the system requirements on the basis of the theoretical design. T is the self-tuning period which should reflect the variation of the harmonic error.

Fig. 9 shows the relationship between the suppressed mean error obtained by fitting experimental data and the iterative learning gains under different load conditions. As can be seen from Fig. 9, when the iterative learning gain self-tuning strategy is used, the suppressed mean error can be maintained at the expected value. At light loads, Γ and Φ are adjusted to smaller values, such as Γ & $\Phi = 0.35$ at $T_e = 25\%$. As the load becomes heavier, Γ and Φ are adjusted to larger values, such as Γ & $\Phi = 0.48$ at $T_e = 100\%$. This strategy improves the suppression ability of the ERPHE, and weakens the amplifying effect of aperiodic disturbance signals.

TABLE I
IPMSM PARAMETERS AND INJECTION INFORMATION

Rated power (kW)	2.2	Rated voltage (V)	380
Rated torque (N·m)	20	Rated current (A)	5.6
Rated frequency (Hz)	50	Phase resistance (Ω)	2.53
Rated speed (r/min)	1000	d-axis inductance (mH)	45
Pairs of poles	3	q-axis inductance (mH)	60
Injected magnitude (V)	62	Injected frequency (Hz)	750

IV. EXPERIMENTAL RESULTS

The proposed method was verified on a 2.2-kW IPMSM experimental platform, as shown in Fig. 10. The test IPMSM and a load IM are coaxially connected. The parameters of test IPMSM and the injection information are listed in Table I. The PWM carrier frequency, the sampling frequency, and the switching frequency are set as 6 kHz. An ARM chip STM32F103VCT6 is applied to execute the entire sensorless control algorithm. An incremental encoder with 2000 p/r is applied to detect the actual rotor position only for comparison. The unsuppressed and the suppressed ERPHE are denoted by $\tilde{\theta}_e$ and $\tilde{\theta}'_e$, respectively.

Fig. 11 shows the experimental results of the influence of Γ and Φ on ERPHE suppression. As can be seen from Fig. 11(a), when Γ and Φ are selected to be smaller values of 0.1, under no-load, 50% rated load, and 100% rated load conditions, the peak-to-peak values (PPVs) of the ERPHE are 6.4°, 7.9°, and 9.5°, respectively. In Fig. 11(b), when Γ and Φ are increased to larger values of 0.3, the ERPHE suppression effect is significantly better than Fig. 11(a). Under the same load condition, the PPVs of the ERPHE are 4.5°, 5.6°, and 6.6°, respectively. In the case of 100% rated load disturbance, the transient position error is 13.1° and 13.2° in Fig. 11(a) and (b), respectively. Furthermore, when Γ and Φ are constant, as the load becomes

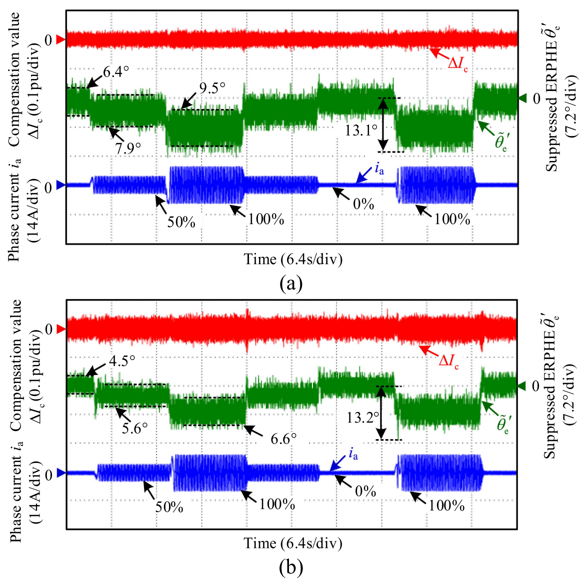

Fig. 11. Experimental results of the influence of Γ and Φ on ERPHE suppression. (a) $\Gamma = \Phi = 0.1$. (b) $\Gamma = \Phi = 0.3$.

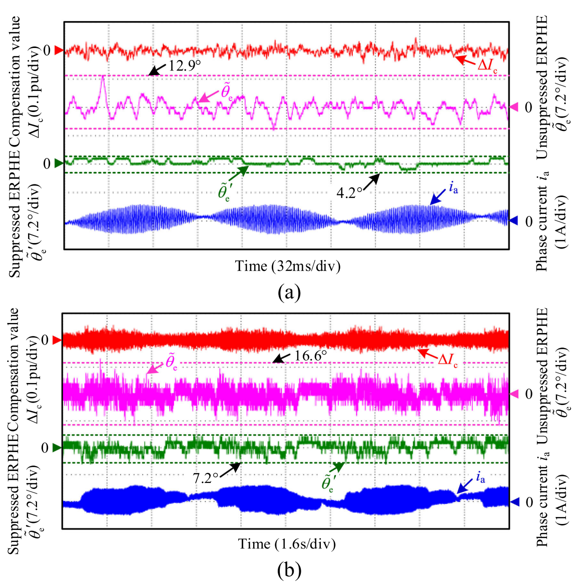

Fig. 12. Experimental results of ERPHE suppression under different speed and no-load conditions. (a) 10% rated speed (100 r/min). (b) 0.2% rated speed (2 r/min).

heavier, the ERPHE also becomes larger, which demonstrates the necessity of the gain self-tuning strategy.

Fig. 12(a) and (b) shows the experimental results of ERPHE suppression under no-load condition at 10% and 0.2% rated speed, respectively. As can be seen from Fig. 12(a), the PPVs of the unsuppressed and the suppressed ERPHE are 12.9° and 4.2°, respectively, which shows an error suppression rate of 67.4%. In Fig. 12(b), the PPVs of the ERPHE are 16.6° and 7.2° before and after using this suppression method, respectively, showing an error suppression rate of 56.6%. This method does not distinguish harmonic error frequencies, which has strong suppression ability for the ERPHE at different speeds.

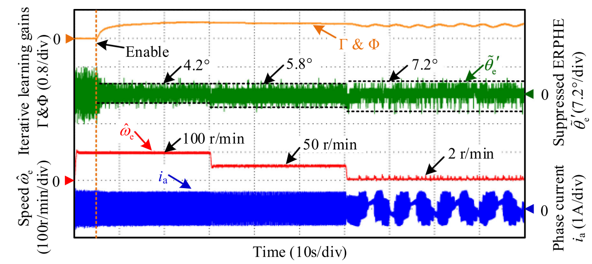

Fig. 13. Experimental results of ERPHE suppression under continuous speed variation conditions.

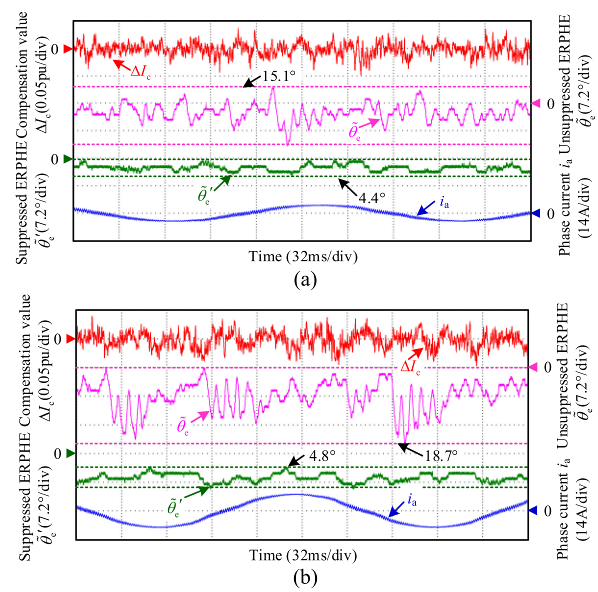

Fig. 14. Experimental results of ERPHE suppression under different load conditions. (a) 50% rated load. (b) 100% rated load.

Fig. 13 shows the experimental results of ERPHE suppression under continuous speed variation conditions. It can be seen that the maximum PPV of the ERPHE is 7.2° when the speed is continuously changed to 100, 50, and 2 r/min. The iterative learning gains Γ and Φ are adaptively adjusted under different speed conditions. When the motor operates at 2 r/min, the tuning range of the learning gains is relatively obvious due to the large fluctuations of the ERPHE.

Fig. 14 shows the experimental results of ERPHE suppression under different load conditions when the motor operates at 10% rated speed (100 r/min). In Fig. 14(a), under 50% rated load condition, the PPVs of the unsuppressed and the suppressed ERPHE are 15.1° and 4.4°, respectively, which shows an error suppression rate of 70.9%. In Fig. 14(b), under 100% rated load condition, the PPVs of the ERPHE are 18.7° and 4.8° before and after using this method, respectively. The error suppression rate is 74.3%. The ERPHE suppression effect is satisfactory.

Fig. 15 shows the FFT analysis results of unsuppressed and suppressed ERPHE under 100% rated load and 10% rated speed (100 r/min, 5 Hz) conditions. In Fig. 15(a), the main harmonic

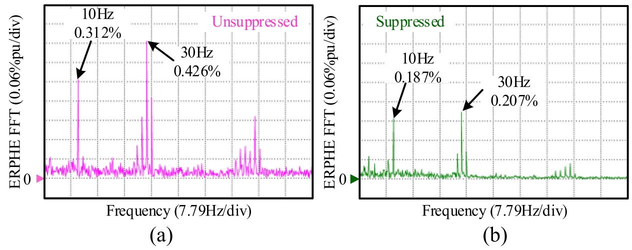

Fig. 15. FFT analysis results of unsuppressed and suppressed ERPHE under 100% rated load and 10% rated speed conditions. (a) FFT analysis of unsuppressed ERPHE. (b) FFT analysis of suppressed ERPHE.

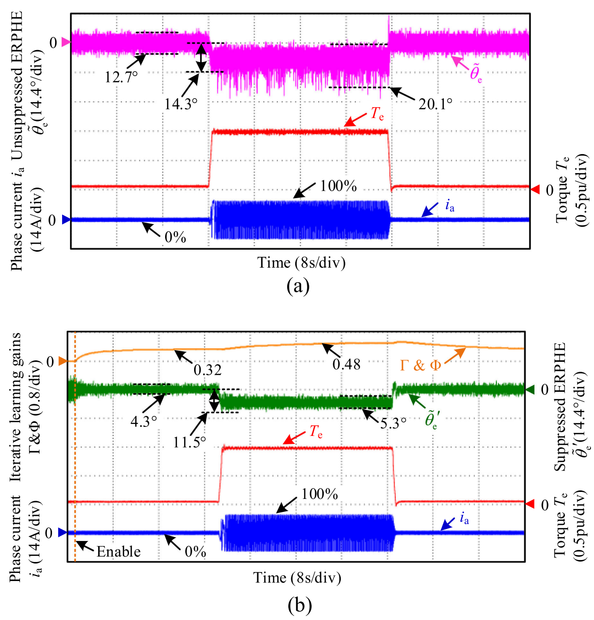

Fig. 16. Experimental results of ERPHE suppression under the condition of 100% rated load disturbance. (a) Without suppression method. (b) With the proposed method.

components of the unsuppressed ERPHE are the second (10 Hz) and the sixth (30 Hz) harmonics, and their amplitude per unit (pu) values are 0.312% and 0.426%, respectively. Moreover, the unsuppressed ERPHE contains a lot of harmonic interference at other frequencies. After using the proposed method, as shown in Fig. 15(b), the second and the sixth harmonic components are reduced to 0.187% and 0.207%, which are suppressed by 40.1% and 51.4%, respectively. It is worth noting that the harmonic interference at other frequencies is also diminished.

The experimental results of ERPHE suppression under the condition of 100% rated load disturbance are shown in Fig. 16. In Fig. 16(a), when the suppression algorithm is not used, the PPVs of the ERPHE reach 12.7° and 20.1° under no-load and 100% rated load conditions, respectively. The transient position error reaches 14.3° in the case of 100% rated load disturbance. After using the proposed method, as shown in Fig. 16(b), the PPVs of the ERPHE are reduced to 4.3° and 5.3° at no load and 100% rated load, respectively. The transient position error is 11.5° and the torque ripple is suppressed. Γ and Φ are also adaptively tuned

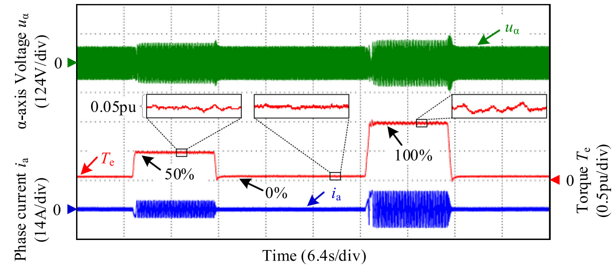

Fig. 17. Experimental results of current, voltage, and torque under 100% rated load, 50% rated load, and no-load conditions.

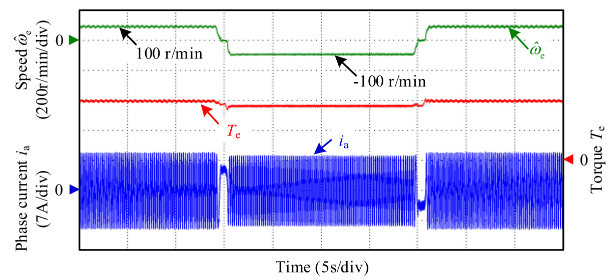

Fig. 18. Experimental result of reversal speed variation with the proposed method under 100% rated load condition.

at different loads to ensure the ERPHE suppression ability, such as 0.32 at no load and 0.48 at 100% rated load.

Fig. 17 shows the experimental results of current, voltage, and torque under 100% rated load, 50% rated load, and no-load conditions. When the proposed suppression method is applied, the ripples of the HF current envelopes, the HF voltage envelopes, and the torque are effectively reduced. The torque ripple is around 0.025 pu at 100% rated load. Furthermore, the motor can operate stably under the load disturbance condition.

Fig. 18 shows the experimental result of reversal speed variation with the proposed method under 100% rated load condition. When the speed changes from 100 r/min to -100 r/min to 100 r/min, the motor can operate smoothly and the ripples of the torque and the speed are relatively small, which further verifies the effectiveness of the proposed method.

Fig. 19 shows the experimental results of ERPHE suppression with the proportional resonant controller, the repetitive controller, and the adaptive frequency filter under 10% rated speed (5 Hz) condition. The resonance frequency of the proportional resonant controller and the center frequency of the adaptive filter are set as sixth harmonic frequency (30 Hz). The digital frequency of the repetitive controller is set as 200. From Fig. 19, these three methods can suppress the ERPHE at different loads. By comparing Figs. 16 and 19, affected by the selection precision of the resonance frequency, the center frequency, and the digital frequency, the suppression performance of these existing methods for complex harmonic error components is limited compared with the proposed method. Furthermore, from the FFT analysis results in Figs. 15 and 19, the suppression ability of these existing methods for the harmonic interference at other frequencies is limited.

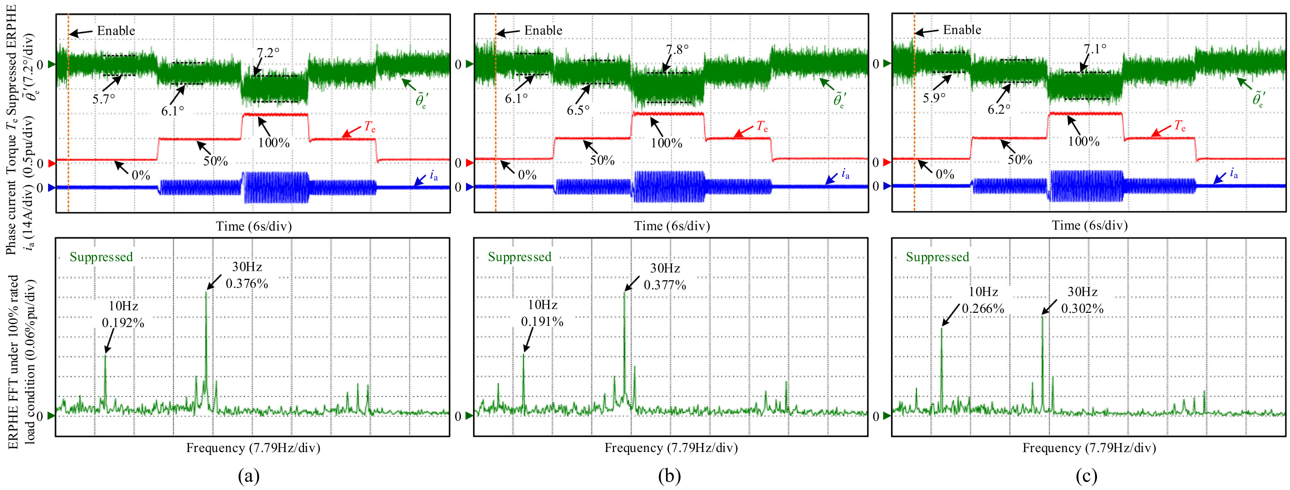

Fig. 19. Experimental results of ERPHE suppression with different methods under 10% rated speed (5 Hz) condition. (a) Proportional resonant controller. (b) Repetitive controller. (c) Adaptive frequency filter.

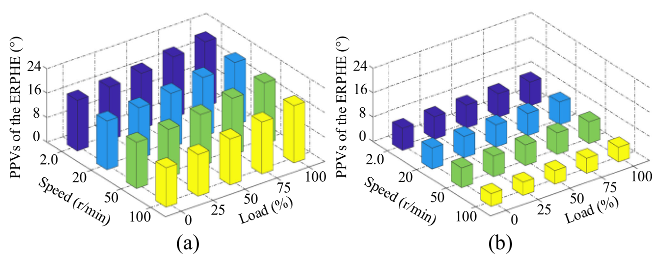

Fig. 20. Experimental comparison results of ERPHE suppression under different speed and load conditions. (a) Without suppression method. (b) With the proposed method.

The experimental comparison results of the ERPHE suppression under different speed and load conditions are shown in Fig. 20. In Fig. 20(a), when the suppression algorithm is not used, the PPVs of the ERPHE reach a maximum of 21.2° at 2 r/min and 100% rated load. When the proposed method is applied, the ERPHE is suppressed effectively and the average error suppression rate reaches more than 50%.

V. Conclusion

This article proposed an adaptive iterative learning control-based online ERPHE suppression method for sensorless PMSM drives. In this method, a PILO with the forgetting factor was proposed, and its stability, convergence, and parameter sensitivity were theoretically analyzed in the discrete time domain. Additionally, an iterative learning gain self-tuning strategy was investigated, improving the ERPHE suppression ability for different loads and speeds. This method does not need to distinguish harmonic error frequencies, avoiding the parameter design of the adaptive filter, the proportional resonant controller, or the repetitive controller. It is independent of the HF injection signal types, and had the advantages of simple structure and good universality. This method provided an effective solution for the practical application requirements of high-performance HFSI-based sensorless drive. The experimental results verified the effectiveness of this method and the average error suppression rate is more than 50%.

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